

FraudSentinel AI

Real-Time Fraud Detection and Prevention in Digital Lending Ecosystems

- Team: [Team Name]
- Value proposition: Stop fraud in milliseconds, explain every decision in plain English.

FraudSentinel AI — Synchrony Hackathon

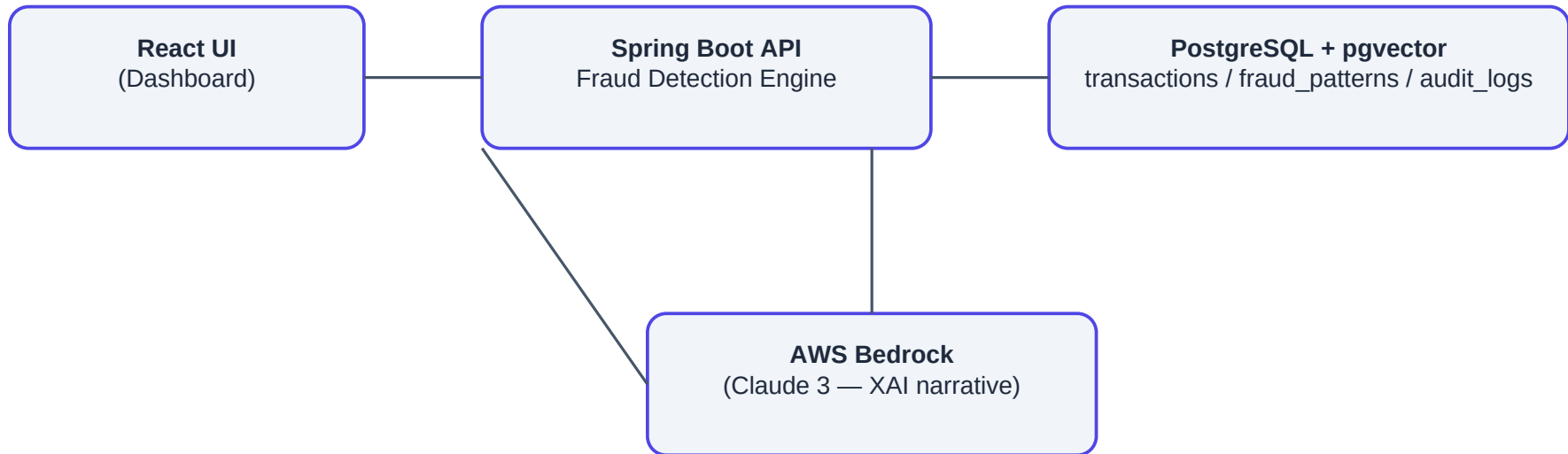
Digital lending fraud is outpacing static defenses

- Fraud is evolving faster than static rule engines can adapt — synthetic identities, loan stacking, device farming, and bust-out schemes now blend into normal-looking behavior.
- Most fraud models today are black boxes. A flagged applicant gets no explanation, compliance teams can't audit the decision, and analysts can't trust or improve the model.
- Speed and trust are in tension: aggressive fraud filters slow down or wrongly block legitimate borrowers, damaging the customer relationship Synchrony depends on.

A hybrid, explainable fraud detection platform

- Deterministic rule checks for known hard signals.
- A real logistic-regression ML fraud-probability score for statistical risk.
- Vector similarity search (pgvector) against a continuously growing library of historical attack signatures.
- AWS Bedrock-powered Explainable AI (XAI) that turns the combined evidence into a plain-English narrative for every decision — Approve, Flag, or Block.

System architecture



- React dashboard — live transaction feed, XAI modal, analyst feedback loop.
- Spring Boot service layer — orchestrates rules, ML scoring, vector search, and XAI in a single `evaluate()` call.
- PostgreSQL with pgvector — HNSW-indexed behavioral embeddings for sub-50ms similarity search.
- AWS Bedrock — guardrailed prompt templates generate the audit-ready explanation.

Rules + ML + pgvector + Bedrock XAI

- Rule engine catches deterministic red flags (high-value transactions, missing device fingerprint, automated-channel loan applications) in under a millisecond.
- The ML score is a real logistic-regression model over engineered features (amount, channel risk, transaction-type risk, device signal) — pluggable for a hosted XGBoost/SageMaker endpoint.
- pgvector cosine similarity search matches the transaction's behavioral embedding against confirmed fraud patterns, surfacing the closest historical attacks and their similarity score.
- Bedrock XAI receives only the computed evidence — never raw customer PII — and returns a structured JSON explanation, never the decision itself.

Guardrails and regulatory readiness

- The LLM explains; it never decides. The approve/flag/block outcome is always computed deterministically before Bedrock is called.
- Prompt templates restrict the model to supplied evidence only, block speculation about identity or protected characteristics, and enforce strict JSON output that is schema-validated before use.
- Every decision writes an immutable audit_logs row — decision, confidence, risk factors, feature importance, and the exact XAI narrative shown to the analyst.
- A Bedrock outage falls back to a deterministic, non-LLM explanation, so the platform never fails open or fails silently.

Built for production from day one

- Stateless Spring Boot service, horizontally scalable behind a load balancer; JWT bearer auth on every API route.
- Zero hardcoded secrets — JWT signing key, DB credentials, and Bedrock model ID are all environment-variable driven.
- HNSW vector indexing keeps similarity search sub-50ms even as the fraud_patterns library grows into the millions of rows.
- Input validation (Bean Validation) and centralized exception handling ensure no stack traces or internal details leak to clients.

What this means for Synchrony

- Sub-200ms end-to-end decision latency, tracked live on the dashboard.
- Every flagged transaction ships with an explanation an analyst can act on in seconds, not minutes.
- The analyst feedback loop (Confirm Fraud / False Positive) immediately promotes confirmed cases into the pattern library, so the system gets sharper with every review.
- Projected impact: reduced false-positive rate, faster analyst throughput, and an audit trail ready for regulatory review.

Where this goes next

- Replace the hand-calibrated logistic-regression weights with coefficients trained on Synchrony's historical lending data.
- Add graph-based entity resolution to catch multi-account and synthetic-identity rings that single-transaction analysis misses.
- Integrate directly into Synchrony's existing loan origination and servicing pipeline via the same REST contract.
- Extend the Bedrock guardrails with a dedicated compliance-review prompt path for SAR drafting.

Thank you

- Live walkthrough: start the transaction stream, watch decisions render in real time.
- Click a flagged transaction to open the XAI modal and read the Bedrock-generated explanation.
- Submit analyst feedback and show the pattern library updating.
- Questions and discussion.

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